

1. If the area of the region $S = \{(x, y) : 2y - y^2 \leq x^2 \leq 2y, x \geq y\}$ is equal to $\frac{n+2}{n+1} - \frac{\pi}{n-1}$, then the natural number n is equal to _____.
[2023 (06 Apr Shift 1)]
2. The area bounded by the curves $y = |x - 1| + |x - 2|$ and $y = 3$ is equal to
[2023 (06 Apr Shift 2)]
(1) 4
(2) 6
(3) 3
(4) 5
3. The area of the region $\{(x, y) : x^2 \leq y \leq 8 - x^2, y \leq 7\}$ is
[2023 (08 Apr Shift 1)]
(1) 27
(2) 18
(3) 20
(4) 21
4. Let the area enclosed by the lines $x + y = 2$, $y = 0$, $x = 0$ and the curve $f(x) = \min\{x^2 + \frac{3}{4}, 1 + [x]\}$ where $[x]$ denotes the greatest integer $\leq x$, be A .
Then the value of $12A$ is
[2023 (08 Apr Shift 2)]
5. Let $y = p(x)$ be the parabola passing through the points $(-1, 0)$, $(0, 1)$ and $(1, 0)$. If the area of the region $\{(x, y) : (x + 1)^2 + (y - 1)^2 \leq 1, y \leq p(x)\}$ is A , then $12(\pi - 4A)$ is equal to _____.
[2023 (10 Apr Shift 1)]
6. If the area of the region $\{(x, y) : |x^2 - 2| \leq y \leq x\}$ is A , then $6A + 16\sqrt{2}$ is equal to _____.
[2023 (10 Apr Shift 2)]
7. Area of the region $\{(x, y) : x^2 + (y - 2)^2 \leq 4, x^2 \geq 2y\}$ is
[2023 (11 Apr Shift 1)]
(1) $\pi + \frac{8}{3}$
(2) $2\pi + \frac{16}{3}$
(3) $\pi - \frac{8}{3}$
(4) $2\pi - \frac{16}{3}$
8. If A is the area in the first quadrant enclosed by the curve $C : 2x^2 - y + 1 = 0$, the tangent to C at the point $(1, 3)$ and the line $x + y = 1$, then the value of $60A$ is.....
[2023 (11 Apr Shift 2)]
9. The area of the region enclosed by the curve $y = x^3$ and its tangent at the point $(-1, -1)$ is
[2023 (12 Apr Shift 1)]
(1) $\frac{19}{4}$
(2) $\frac{23}{4}$
(3) $\frac{31}{4}$
(4) $\frac{27}{4}$
10. The area of the region enclosed by the curve $f(x) = \max\{\sin x, \cos x\}$, $-\pi \leq x \leq \pi$ and the x -axis is
[2023 (13 Apr Shift 1)]
(1) $2\sqrt{2}(\sqrt{2} + 1)$
(2) 4
(3) $4(\sqrt{2})$
(4) $2(\sqrt{2} + 1)$
11. The area of the region $x, y : x^2 \leq y \leq |x^2 - 4|, y \geq 1$ is
[2023 (13 Apr Shift 2)]
(1) $\frac{4}{3}(4\sqrt{2} - 1)$
(2) $\frac{4}{3}(4\sqrt{2} + 1)$
(3) $\frac{3}{4}(4\sqrt{2} + 1)$
(4) $\frac{3}{4}(4\sqrt{2} - 1)$
12. If the area bounded by the curve $2y^2 = 3x$, lines $x + y = 3$, $y = 0$ and outside the circle $(x - 3)^2 + y^2 = 2$ is A , then $4(\pi + 4A)$ is equal to _____.
[2023 (15 Apr Shift 1)]

ANSWER KEYS

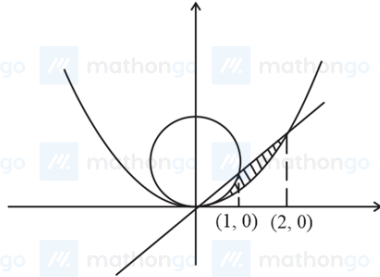
1. (5) 2. (1) 3. (3) 4. (17) 5. (16) 6. (27) 7. (4) 8. (16)
9. (4) 10. (2) 11. (1) 12. (42)

1. (5)

Given,

$$S = \{(x, y) : 2y - y^2 \leq x^2 \leq 2y, x \geq y\},$$

Now plotting the diagram, we get



Now from above diagram, given region are,

$$2y - y^2 \leq x^2, x^2 \leq 2y \text{ and } x \geq y$$

Now the intersecting point of $x^2 + y^2 - 2y = 0$ & $x = y$ will be $(1, 1)$ and $x^2 = 2y$ & $x = y$ will be $(2, 2)$

So, required area will be,

$$A = \int_0^1 \left(1 + \sqrt{1 - x^2} - \frac{x^2}{2}\right) dx + \int_1^2 \left(x - \frac{x^2}{2}\right) dx$$

$$\Rightarrow A = \left(x - \frac{x^2}{2} \sqrt{1 - x^2} - \frac{1}{2} \cdot \sin^{-1} x - \frac{x^3}{6}\right)_0^1 + \left(\frac{x^2}{2} - \frac{x^3}{6}\right)_1^2$$

$$\Rightarrow A = \frac{7}{6} - \frac{\pi}{4} = \frac{n+2}{n+1} - \frac{\pi}{x-1}$$

So, on comparing we get, $n = 5$

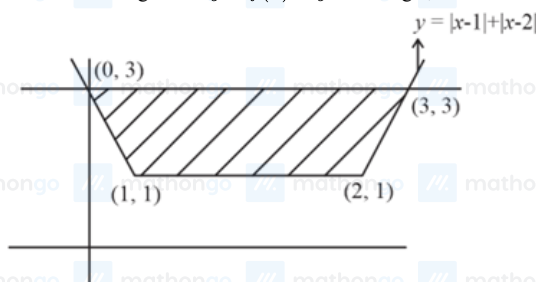
2. (1)

Given:

$$y = f(x) = |x - 1| + |x - 2|$$

$$\Rightarrow f(x) = \begin{cases} 3 - 2x & : x < 1 \\ 1 & : 1 \leq x < 2 \\ 2x - 3 & : x \geq 2 \end{cases}$$

Let us draw diagram of $y = f(x)$ & $y = 3$ we get,



Required area is the area of trapezium which is

$$\text{Area} = \frac{1}{2} [1 + 3] \times 2$$

$$= 4 \text{ sq. units}$$

3. (3)

The given curves are $x^2 \leq y$, $y \leq 8 - x^2$ and $y \leq 7$

The point of intersection of the curves $x^2 \leq y$ and $y \leq 8 - x^2$ is obtained by,
 $\Rightarrow x^2 = 8 - x^2$

$\Rightarrow x = \pm 2$ and $y = 4$.

Hence the points are $(2, 4), (-2, 4)$.

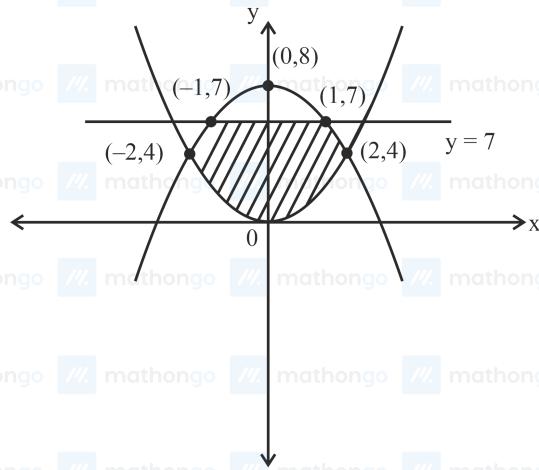
The points of intersection of the curves $y \leq 8 - x^2$ and $y \leq 7$ is obtained by,

$\Rightarrow 8 - x^2 = 7$

$\Rightarrow x = \pm 1$ and $y = 7$

Hence the points are $(1, 7), (-1, 7)$.

The required graph is



The required area is symmetrical about y -axis.

Hence required area is $A = 2 \left[\int_0^4 \sqrt{y} \, dy + \int_4^7 \sqrt{8-y} \, dy \right]$

$$A = 2 \left[\left[\frac{y^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^4 - \left[\frac{(8-y)^{\frac{3}{2}}}{\frac{3}{2}} \right]_4^7 \right]$$

$$= \frac{4}{3} \left[(4)^{\frac{3}{2}} - (0)^{\frac{3}{2}} - \left\{ (8-7)^{\frac{3}{2}} - (8-4)^{\frac{3}{2}} \right\} \right]$$

$$= \frac{4}{3} (8 - 0 - 1 + 8)$$

$$= 20 \text{ sq units.}$$

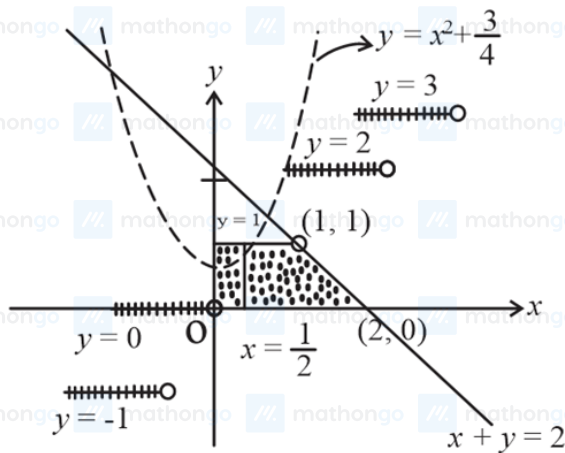
Hence, the required area is 20 sq units.

4. (17)

Given,

The lines $x + y = 2$, $y = 0$, $x = 0$ and the curve $f(x) = \min\left\{x^2 + \frac{3}{4}, 1 + [x]\right\}$ where $[x]$ denotes the greatest integer $\leq x$,

Now plotting the diagram of given function we get,



Now from above diagram, the area enclosed is given by,

$$A = \left[\int_0^{\frac{1}{2}} \left(x^2 + \frac{3}{4} \right) dx \right] + \frac{1}{2} \left(\frac{1}{2} + \frac{3}{2} \right) \times 1 = \frac{5}{12} + 1$$

$$\Rightarrow 12A = 17$$

5. (16)

Let, the parabola be $y = ax^2 + bx + c$,

Now given parabola is passing through $(-1, 0)$, $(0, 1)$ & $(1, 0)$

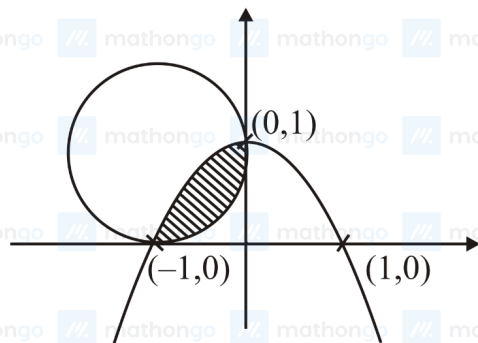
Now putting the value in the above equation of parabola we get,

$$a - b + c = 0 \dots (1), c = 1 \dots (2) \text{ and } a + b + c = 0 \dots (3)$$

Now solving above three equations we get,

$$\text{Parabola as } y = 1 - x^2$$

Now area of region between $y = 1 - x^2$ and $(x + 1)^2 + (y - 1)^2 \leq 1$ is given by,



Now from above diagram, required area will be,

$$A = \int_{-1}^0 \left(-x^2 + \sqrt{1 - (x + 1)^2} \right) dx$$

$$\Rightarrow A = \frac{\pi}{4} - \frac{1}{3}$$

$$\text{Hence, } 12(\pi - 4A) = 16$$

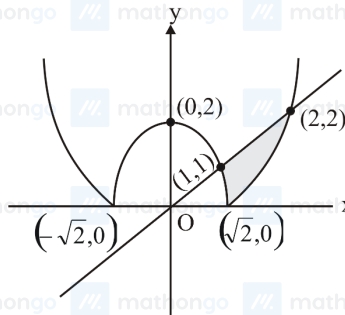
6. (27)

Given,

$$|x^2 - 2| \leq y \text{ \& } y \leq x$$

Now on finding the intersection point we get, $(x, y) \equiv (1, 1) \text{ \& } (2, 2)$

Now plotting the curve of the above function we get,



Now from above diagram required bounded area will be,

$$A = \left[\int_1^{\sqrt{2}} (x - (x^2 - 2)) dx + \int_{\sqrt{2}}^2 (x - (x^2 - 2)) dx \right]$$

$$\Rightarrow A = \left[\frac{x^2}{2} - 2x + \frac{x^3}{3} \right]_1^{\sqrt{2}} + \left[\frac{x^2}{2} - \frac{x^3}{3} + 2x \right]_{\sqrt{2}}^2$$

$$\Rightarrow A = \left[1 - 2\sqrt{2} + \frac{2\sqrt{2}}{3} \right] - \left[\frac{1}{2} - 2 + \frac{1}{3} \right] + \left[2 - \frac{8}{3} + 4 \right] - \left[1 - \frac{2\sqrt{2}}{3} + 2\sqrt{2} \right]$$

$$\Rightarrow A = \frac{9}{2} - \frac{8\sqrt{2}}{3}$$

$$\Rightarrow 6A = 27 - 16\sqrt{2}$$

$$\Rightarrow 6A + 16\sqrt{2} = 27$$

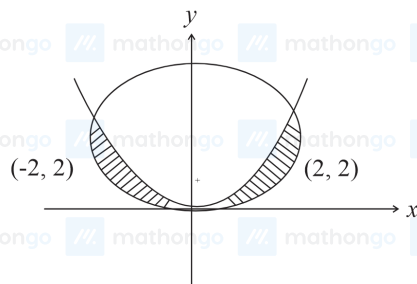
7. (4)

Given,

$$x^2 + (y - 2)^2 \leq 4 \text{ \& } x^2 \geq 2y$$

Now on solving, $x^2 + (y - 2)^2 = 4$ \& $x^2 = 2y$ we get intersecting point of the curves as $(\pm 2, 2)$,

Now plotting the diagram we get,



So, from above diagram area of required region will be,

$$= 2 \left[\frac{1}{4} (\pi \times 2^2) - \int_0^2 \sqrt{2} \cdot \sqrt{y} dy \right]$$

$$= 2 \left[\pi - \left[\frac{\sqrt{2} y^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^2 \right]$$

$$= 2 \left[\pi - \frac{2\sqrt{2}}{3} \cdot 2\sqrt{2} \right]$$

$$= 2 \left[\pi - \frac{8}{3} \right]$$

$$= 2\pi - \frac{16}{3}$$

8. (16)

Given,

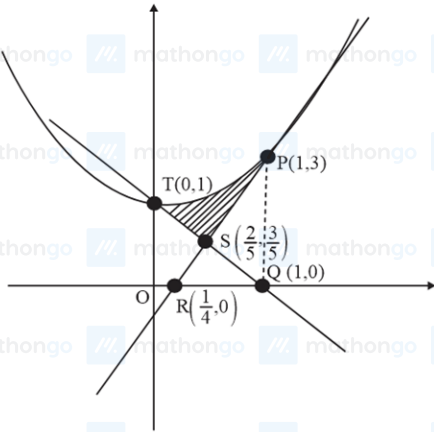
$$y = 2x^2 + 1$$

Now tangent to above equation at (1, 3) is given by,

$$y - 3 = 4(x - 1)$$

$$\Rightarrow y = 4x - 1$$

Now plotting the diagram we get,



Required area of bounded region will be,

$$A = \int_0^1 (2x^2 + 1) dx - \text{area of } \triangle QOT - \text{area of } \triangle PQR + \text{area of } \triangle QRS$$

$$\Rightarrow A = \left(\frac{2}{3} + 1\right) - \frac{1}{2} - \frac{9}{8} + \frac{9}{40} = \frac{16}{40}$$

$$\Rightarrow 60A = 16$$

9. (4)

Given $y = x^3$... (i)

$$\Rightarrow \frac{dy}{dx} = 3x^2$$

$$\left(\frac{dy}{dx}\right)_{(-1, -1)} = 3$$

Equation of tangent at $(-1, -1)$

$$(y + 1) = 3(x + 1)$$

$$y = 3x + 2 \quad \dots (ii)$$

Solving (i) and (ii)

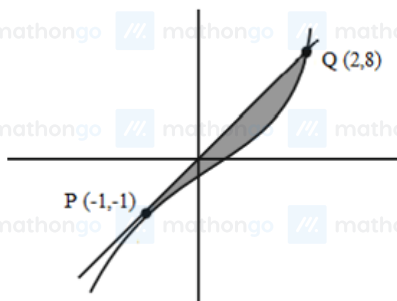
$$\Rightarrow x^3 = 3x + 2$$

$$\Rightarrow x^3 - 3x + 2 = 0$$

$$\Rightarrow x = -1, -1, 2$$

Another point of intersection is $Q(2, 8)$

So, now plotting the diagram we get,



Required area from the above diagram will be,

$$= \int_{-1}^2 (3x + 2 - x^3) dx$$

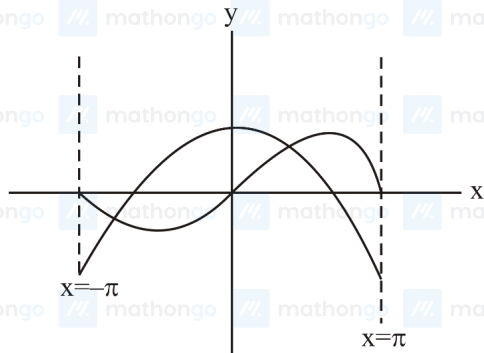
$$= \frac{3}{2}(4 - 1) + 2(2 + 1) - \frac{1}{4}(16 - 1)$$

$$= \frac{27}{4}$$

Hence this is the required option.

10. (2)

Plotting the required diagram we get,



Now from diagram we can say that,

$\sin x$ is maximum for $x \in \left(-\pi, -\frac{3\pi}{4}\right) \cup \left(\frac{\pi}{4}, \pi\right)$ and $\cos x$ is maximum for $x \in \left(-\frac{3\pi}{4}, -\frac{\pi}{2}\right) \cup \left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$

So, area of the region is given by,

$$A = \left| \int_{-\pi}^{-\frac{3\pi}{4}} \sin x dx \right| + \left| \int_{-\frac{3\pi}{4}}^{-\frac{\pi}{2}} \sin x dx \right| + \left| \int_{-\frac{\pi}{2}}^{-\frac{\pi}{4}} \cos x dx \right| + \left| \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos x dx \right| + \left| \int_{\frac{\pi}{4}}^{\pi} \sin x dx \right|$$

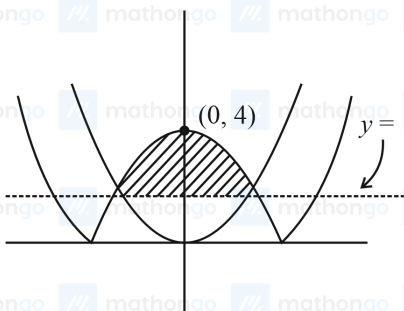
$$\Rightarrow A = \left| \frac{1}{\sqrt{2}} - 1 \right| + \left| 1 + \frac{1}{\sqrt{2}} \right| + \left| -1 + \frac{1}{\sqrt{2}} \right| + \left| \frac{1}{\sqrt{2}} + 1 \right|$$

$$\Rightarrow A = 2 + \frac{2}{\sqrt{2}} + 2 - \frac{2}{\sqrt{2}} = 4$$

Hence this is the required option.

11. (1)

The required diagram is:



The shaded region is symmetric about y -axis.

$$\text{Required area} = 2 \left[\int_1^2 \sqrt{y} dy + \int_2^4 \sqrt{4-y} dy \right]$$

$$= 2 \left[\frac{y^{3/2}}{3/2} \right]_1^2 - \left[\frac{2(4-y)^{3/2}}{3} \right]_2^4$$

$$= \frac{4}{3} (4\sqrt{2} - 1)$$

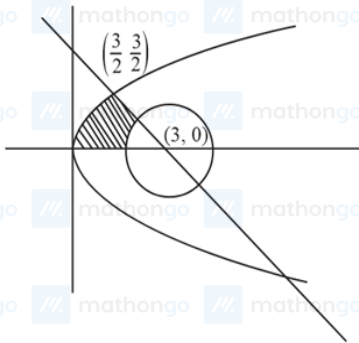
Hence this is the correct option.

12. (42)

Given:

$$2y^2 = 3x, x + y = 3 (\text{slope is } -1) \text{ \& } (x-3)^2 + y^2 = 2$$

Now plotting the diagram of the above function we have,



So, required area is

$$A = \int_0^{\frac{3}{2}} \left(3 - y - \frac{2y^2}{3} \right) dy - \frac{\pi \left(\sqrt{2} \right)^2 \times 45^\circ}{360^\circ}$$

$$\Rightarrow A = \left[3y - \frac{y^2}{2} \right]_0^{\frac{3}{2}} - \frac{2}{3} \left[\frac{y^3}{3} \right]_0^{\frac{3}{2}} - \frac{\pi}{4}$$

$$\Rightarrow A = \frac{21}{8} - \frac{\pi}{4}$$

So,

$$A = \frac{1}{4} \left(\frac{21}{2} - \pi \right)$$

$$\Rightarrow \pi + 4A = \frac{21}{2}$$

$$\Rightarrow 4(\pi + 4A) = 42$$